

Xue (Snow) Yu

Stanford University | Division of Nephrology, School of Medicine
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EDUCATION

Georgetown University

2017–2019

Master of Science, Biostatistics | Washington, DC

Master's research project: Statistical Tests for the Comparison of Several Medians. Advisor: George Luta, PhD.

Hong Kong Baptist University

2013–2017

Bachelor of Science (Honours), Financial Mathematics | HongKong

RESEARCH INTERESTS

Statistical and computational methods for kidney and population health; longitudinal modeling; model evaluation and population representativeness; reproducible analysis; responsible evaluation of AI in healthcare.

RESEARCH EXPERIENCE

Stanford University

Apr 2022–Present

Biostatistician II | Division of Nephrology, School of Medicine

1. Compared OLS, linear mixed-effects, and latent class mixed-effects models in R to identify rapid kidney function decline in 292 Sri Lankan participants. Conducted pooled and sex-stratified analyses and wrote the first-author JSM 2024 poster; jointly planned and validated analyses and interpreted classification differences and censoring limitations with a faculty mentor.
2. Computed demographic representativeness scores and programmed comparisons of RADx-UP with 2020 U.S. Census data to identify demographic gaps obscured by aggregate summaries. Wrote the first-author JSM 2025 poster; jointly developed the analysis plan, validated analyses, and interpreted findings with a faculty mentor.
3. Developed reusable RADx-UP trial data workflows and R/SAS functions for baseline tables, standardized differences, and missing-data summaries.
4. Designed and launched a web-based flowchart tool for fellows without coding experience, supporting multiple export formats and recording over 200 uses from June 2025 to January 2026.

University of Mississippi Medical Center

Jun 2019–Feb 2022

Biostatistician; promoted to Biostatistician II in 2022

Department of Data Science, Bower School of Population Health

5. First author of a QTc study in 293 adults and 121 children with sickle cell disease using multivariable regression and a published Jackson Heart Study model; contributed with J. F. Maher to a sensitivity analysis excluding six adult diuretic users.
6. Analyzed metabolic changes during sickle cell pain crises; coauthored an amino-acid study using partial least squares-discriminant analysis and genetic research on QT prolongation.

RESEARCH EXPERIENCE CONTINUED

Georgetown University

Jan–Dec 2018

Research Assistant | Department of Biostatistics

7. Prepared data, programmed statistical analyses, and contributed to analysis plans for clinical and laboratory studies.

JOURNAL PUBLICATIONS

First-author publication

8. **Yu X**, Majumdar S, Pollard JD, Jackson E, Knudson J, Wolfe D, Kato GJ, Maher JF. Clinical and Laboratory Correlates of QTc Duration in Adult and Pediatric Sickle Cell Disease. *American Journal of Medicine Open*. 2023;10:100045. doi:10.1016/j.ajmo.2023.100045

Coauthored publications

9. Elliott A, Anand S, Heller HC, Orellana E, Contreras Nieves M, Ley C, Montez Rath M, **Yu X**, Beville D, Flores L, Casa D, Chicas R, Hertzberg VS, Riden H, Fathallah FA, Glaser J, Morrissey-Basler MC, Parsonnet J. Beyond Water, Rest, and Shade: Advancing Farmworker Heat Protection Through Partnership. *American Journal of Industrial Medicine*. 2026;69(9):687-695. doi:10.1002/ajim.70105
10. Varkila MRJ, Montez-Rath M, **Yu X**, Subramanian N, Owens DK, Brady B, Block GA, Parsonnet J, Chertow GM, Anand S. Dialysis Facility Closures in the US From 2018 to 2024: A Serial Cross-Sectional Study. *American Journal of Kidney Diseases*. 2026;87(6):750-760.e1. doi:10.1053/j.ajkd.2025.12.003
11. Contreras Nieves M, Hewapathirana S, Zabel FQ, Mahalekam D, **Yu X**, Weerakoon C, Bandara M, Bandara I, Maduwantha I, Hearth D, Athauda I, Rathnayake B, Erandika N, Pushpakumara N, Lee E, Baron K, Bhalla V, Artilles KL, Fire A, Schensul SL, Chandrajith R, Vlahos P, Montez-Rath M, Anand S, Nanayakkara N. Case-control study of water and agricultural exposures among women in CKDu-endemic Sri Lanka. *Journal of Water and Health*. 2026;24(2):160-175. doi:10.2166/wh.2026.175
12. Contreras Nieves M, Aziz A, Powers A, Banda E, Jaskulak S, Cerezo M, Ramirez I, **Yu X**, Moreno M, Jimenez D, Ajang I, Garcia P, Dhingra H, Krishna GG, Lunn MR, Kurella Tamura M, Montez-Rath M, Rodriguez J, Yang T, Anand S. Why Am I on Dialysis? Gaps in Rural CKD Care in Central California. *Clinical Journal of the American Society of Nephrology*. 2026;21(9):1660-1663. doi:10.2215/CJN.0000000965
13. Subramanian N, Anand S, **Yu X**, Montez-Rath ME, Jarhyan P, Rajan S, Venkateshmurthy NS, Levin A, Prabhakaran D, Craig P, Prabhakaran P, Mohan S, Jaacks L. Community-Specific Differences in Kidney Function in Rural and Urban India. *Kidney International Reports*. 2025;10(6):2035-2040. doi:10.1016/j.ekir.2025.04.006
14. Montez-Rath M, Varkila M, **Yu X**, Brillhart S, Morgan C, Leppink A, Block MS, Mehta S, Hunsader P, Fountaine A, Subramanian N, Dittrich M, Owens DK, Chertow GM, Parsonnet J, Anand S, Block GA. Acceptance of SARS-CoV-2 Surveillance Testing Among Patients Receiving Dialysis: A Cluster Randomized Trial. *JAMA Network Open*. 2024;7(9):e2434159. doi:10.1001/jamanetworkopen.2024.34159

JOURNAL PUBLICATIONS CONTINUED

15. Thule J, **Yu X**, Montez-Rath ME, Chertow GM. MyPlate Awareness and Engagement and Perceived and Objective Diet Quality in US Adults With Chronic Kidney Disease. *Journal of Renal Nutrition*. 2024;34(6):493-499. doi:10.1053/j.jrn.2024.04.007
16. Zhang M, Hillegass WB, **Yu X**, Majumdar S, Pollard JD, Jackson E, Knudson J, Wolfe D, Kato GJ, Maher JF, Mei H. Genetic variants and effect modifiers of QT interval prolongation in patients with sickle cell disease. *Gene*. 2024;890:147824. doi:10.1016/j.gene.2023.147824
17. Varkila MRJ, Montez-Rath ME, Salomon JA, **Yu X**, Block GA, Owens DK, Chertow GM, Parsonnet J, Anand S. Use of Wastewater Metrics to Track COVID-19 in the US. *JAMA Network Open*. 2023;6(7):e2325591. doi:10.1001/jamanetworkopen.2023.25591
18. Hewavitharana P, Schensul S, Lee E, Montez-Rath M, Senarathne S, Liu S, Harold K, Hewapathirana S, Erandika N, Abeysundara HTK, **Yu X**, Bhalla V, Fire A, Levin A, Anand S, Vlahos P, Chandrajith R, Nanayakkara N. Describing Natural History and Exploring Risk Factors for Kidney Function Decline in Persons With CKD of Uncertain Etiology in Sri Lanka. *Kidney International Reports*. 2023;8(7):1430-1438. doi:10.1016/j.ekir.2023.04.010
19. Anand S, Montez-Rath M, Varkila M, **Yu X**, Block M, Brillhart S, Leppink A, Hunsader P, Owens DK, Chertow GM, Parsonnet J, Block GA. Feasibility and Acceptability of SARS-CoV-2 Screening among Patients Receiving Hemodialysis: A Pilot Study. *Clinical Journal of the American Society of Nephrology*. 2023;18(7):930-932. doi:10.2215/CJN.0000000000000137
20. Zhou Y, **Yu X**, Nicely A, Cunningham G, Challa C, McKinley K, Nickel R, Campbell A, Darbari D, Summar M, Majumdar S. Amino acid signature during sickle cell pain crisis shows significant alterations related to nitric oxide and energy metabolism. *Molecular Genetics and Metabolism*. 2022;137(1-2):146-152. doi:10.1016/j.ymgme.2022.08.004

PREPRINT

Subramanian N, **Yu X**, Varkila M, Hunsader P, Morgan C, Mehta S, Dittrich M, Block G, Owens DK, Parsonnet J, Chertow GM, Anand S, Montez-Rath M. Time of Day of SARS-CoV-2 Vaccination and Antibody Response in Patients Receiving Dialysis. *SSRN*. Preprint posted July 23, 2025. doi:10.2139/ssrn.5337673

FIRST AUTHOR RESEARCH POSTERS

Yu X, Varkila MRJ, Subramanian N, Parsonnet J, Chertow G, Anand S, Montez-Rath M. Demographic Representativeness of the RADx-Up Cohort Compared to the U.S. Census Data. **Joint Statistical Meetings, 2025**.

Also presented at Stanford SRP Poster Symposium (2025), retitled Generalizability Score to Assess Sample Representativeness: Computation and Visualization following JSM feedback.

Yu X, Charu V, Anand S, Nanayakkara N, Vlahos P, Chandrajith R, Montez-Rath M. Comparing statistical models for identifying rapid decline in chronic kidney disease. **Joint Statistical Meetings, 2024**.

Also presented at Stanford SRP Poster Symposium.

MENTORING AND STATISTICAL CONSULTING

21. Mentored undergraduate researchers from the National University of Singapore and master's students from UC San Diego and Stanford Epidemiology.
22. Mentored student research on MyPlate awareness and diet quality in CKD, resulting in a coauthored Journal of Renal Nutrition article; also mentored work on AI prediction of CKDu progression reported in the WCN 2026 abstract.
23. Provided twice-monthly biostatistical office hours beginning in October 2022, and statistical consulting to Q2BI and clinical and statistical teams in Nanjing, China.

LEADERSHIP AND SERVICE

Quantitative Staff Network

Nov 2024–Present

Chair, Communications Subcommittee

24. Led newsletter development, LinkedIn outreach, survey design, member engagement, and mentorship program planning; helped organize career workshops and networking events.

Biostats4You: Reviewer of biostatistics educational resources (2023). Reviewer listing

SELECTED EDUCATIONAL PRESENTATIONS

2025: Cosmos: A Collaborative Platform Empowering Healthcare. Kidney Clinical Research Conference, Stanford (September 25).

2022: Missing Data and Multiple Imputation method. Nephrology Biostatistics group meeting (September 27); Variable selection in multivariable models. Kidney Clinical Research Conference (June 27), Stanford.

COAUTHORED CONFERENCE ABSTRACTS AND POSTERS

Published and presented under the names Xue Yu and Snow Yu. Original author names are retained.

25. Ashktorab Y, Hewapathirana S, **Yu X**, Anand S, Chandrajith R, Nanyakkara N, Montez-Rath M. From algorithms to agriculture: Applying AI methods to predict kidney disease in farmworkers. *Kidney International Reports*. 2026;11:104327. World Congress of Nephrology 2026, abstract WCN26-3861.
26. Contreras Nieves M, Hewapathirana S, Zabel FQ, **Yu X**, Lee E, Baron K, Vlahos P, Chandrajith R, Montez Rath M, Anand S, Nanayakkara N. Case-control study characterizing drinking water and agricultural practices among women in kidney disease endemic and non-endemic regions in Sri Lanka. World Congress of Nephrology 2026. Poster WCN26-3918. Conference record
27. Contreras Nieves M, Hewapathirana S, Zabel FQ, **Yu X**, Baron KD, Vlahos P, Montez-Rath ME, Anand S, Nanayakkara N. Zooming In on the Outliers: A Case-Control Study Characterizing Drinking Water and Agricultural Practices Among Women in Kidney Disease Endemic and Nonendemic Regions in Sri Lanka. *ASN Kidney Week 2025*. Abstract TH-PO1029. doi:10.1681/ASN.2025j31m5hfw
28. Subramanian N, **Yu X**, Varkila M, Block GA, Parsonnet J, Chertow GM, Anand S, Montez-Rath ME. Vaccination Strategies in Patients Receiving Dialysis: Should We Watch the Clock? *ASN Kidney Week 2024*. Abstract TH-PO1149. doi:10.1681/ASN.2024swtrr2jc
29. Varkila M, Montez-Rath ME, **Yu X**, Subramanian N, Chertow GM, Parsonnet J, Anand S. Trends in Dialysis Facility Access in the United States from 2018–2023 and Association with Area Socioeconomic Disadvantage. *ASN Kidney Week 2024*. Abstract TH-OR05. doi:10.1681/ASN.2024ak9m2bac
30. Contreras Nieves M, Powers AJ, Jaskulak-Gonzalez SA, Banda E, **Yu X**, Kurella Tamura M, Anand S. Why Am I on Dialysis? Exploring the Gap between Patient Perspectives and Clinical Diagnosis of Kidney Disease in California’s Central Valley. *ASN Kidney Week 2024*. Abstract FR-PO1034. doi:10.1681/ASN.2024axxdzryb
31. Subramanian N, Prabhakaran P, Venkateshmurthy NS, Jaacks LM, Mohan S, Jarhyan P, Craig P, **Yu X**, Prabhakaran D, Montez-Rath M, Anand S. Kidney function within Indian populations: variation by geography and population-density. World Congress of Nephrology 2024. Poster associated with abstract WCN24-2024. Conference record
32. Montez-Rath ME, Varkila M, **Yu X**, Parsonnet J, Chertow G, Block GA, Anand S. SARS-CoV-2 Testing During Routine Hemodialysis Care: A Nationwide Pragmatic Clinical Trial. *ASN Kidney Week 2023*. Abstract TH-PO1132. Conference record
33. Hewapathirana S, Lee E, Weerakoon C, Erandika N, Herath D, Bandara I, Pushpakumara N, Bandara M, Harold K, **Yu X**, Montez Rath M, Vlahos P, Chandrajith R, Anand S, Nannayakkara N. Approach to early diagnosis of CKDu by assessing an at risk community in Sri Lanka. *Kidney International Reports*. 2023;8(3):S125. World Congress of Nephrology 2023, abstract WCN23-0956. doi:10.1016/j.ekir.2023.02.283

COAUTHORED CONFERENCE ABSTRACTS AND POSTERS CONTINUED

34. Noor R, Akam-Venkata J, Joseph M, Knudson J, Lirette ST, **Yu S**, Majumdar S, Shakti D. Abnormal left ventricular strain in children with sickle cell disease. Poster presentation, American Society of Echocardiography Scientific Sessions. June 10–13, 2022; Seattle, WA. Conference record
35. Tio M, **Yu X**, Hall ME, Shafi T. A comparison of trends in chronic kidney disease in Mississippi using the Chronic Kidney Disease Epidemiology Collaboration 2009 vs. 2021 equations. Southern Regional Meeting. February 10–12, 2022; New Orleans, LA. Listed in two sessions as abstracts 394 and 581. Conference record

TECHNICAL SKILLS

Programming and reproducible research: R, SAS, Stata, Python, SQL, R Markdown, R Shiny, Git, LaTeX. **Methods:** Regression, mixed-effects models, survival analysis, GEE, predictive modeling, missing-data imputation. **Research systems:** Epic Cosmos, REDCap, Qualtrics, Tableau.

SELECTED COMPLETED COURSEWORK

Graduate degree: Machine Learning for Bioinformatics; Data Science; Statistical Inference; Survival Analysis; Linear Models and Multivariate Analysis; Categorical Data Analysis; Experimental Design and Clinical Trials.

Undergraduate degree: Data Structure (for FM Students); C++ Programming Language; Advanced Mathematics I and II; Advanced Probability; Linear Algebra I and II; Stochastic Processes and Applications.

PROFESSIONAL TRAINING AND CREDENTIALS

Audited / non-credit Stanford coursework: BIOMEDIN/BMDS 223, Deploying and Evaluating Fair AI in Healthcare (Spring 2026); EPI 262, Intermediate Biostatistics: Regression, Prediction, and Survival Analysis (Spring 2024).

Professional development: Team Science Principles for Data Scientists, Statistical Horizons (September 2023); LLMs for Statisticians and Data Scientists; Fundamentals of AI/ML in Precision Medicine; Spectrum/Stanford Essentials of Clinical Research (January–March 2025).

SAS: SAS Certified Base Programmer for SAS 9 (April 2019).

Epic: Cosmos Data Model Accreditation (March 2025); Cosmos Researcher Badge (May 2026); Cosmos Super User Badge (August 2026).

HONORS AND AWARDS

National Endeavor Scholarship, Government of China (2016); Annual Merit Award for Student Ambassador, Hong Kong Baptist University (2015).

ADDITIONAL CONFERENCE RECORD TO VERIFY

End-stage kidney disease of unknown etiology: Studying a hotspot in California's Central Valley. Annual Dialysis Conference (2023), adult poster listing 18194. Official poster listing